

Engineering Physics (2025)
Course code 25PY101
Module 2 Unit 1: Quantum theories of solids

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November 26, 2025

Unit 2 Plan

- 1 Quantum Free Electron Theory
- 2 Fermi-Dirac distribution
- 3 Electronic specific heat of solids
- 4 Density of states (qualitative)
- 5 Success and Failures of quantum free electron theory of solids
- 6 E-k diagram

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Specific heat capacity

Definition

- Specific heat capacity ^a C is defined as the rate of change of energy E with temperature T per unit mole of material.

$$C = \frac{dE}{dT} \quad \text{SI unit} \quad [C] = \left[\frac{E}{T} \cdot \frac{1}{\text{mole}} \right] = \text{J K}^{-1} \text{mol}^{-1}.$$

^aIt is also called simply as specific heat.

- Metals heat “quickly”. When we heat a metal, the heat energy is transferred to electrons. The quickness is measured in terms of heat capacity.

Problem

Specific heat of mercury is $0.14 \text{ J g}^{-1} \text{ K}^{-1}$, water is $1 \text{ cal g}^{-1} \text{ }^{\circ}\text{C}^{-1}$, ethanol is $2.44 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$. Which has more specific heat? Which better material for thermometer?

[$^{200}_{80}\text{Hg}$, $1 \text{ cal} = 4.2 \text{ J}$, ethanol = $\text{CH}_3\text{CH}_2\text{OH}$]

Specific heat capacity – CFET

- Energy of electron at temperature T is $E_{\text{el}} = \frac{3}{2}k_B T$.
- Energy of a mole of electrons is

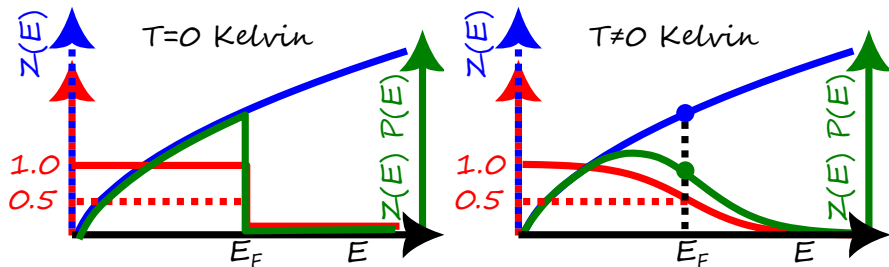
$$E = E_{\text{el}}N_A = \frac{3}{2}k_B N_A T = \frac{3}{2}RT. \quad [R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}]$$

- Therefore, specific heat capacity is

$$C_{\text{theory}}^{\text{classical}} = \frac{dE}{dT} = \frac{3}{2}R \sim 12 \text{ J mol}^{-1} \text{ K}^{-1}.$$

- But experimental values of specific heat capacity $C_{\text{experiment}}$ are **smaller by two orders** of magnitude.

Specific heat capacity – QFET



- For the derivation of specific heat capacity, only the **form** of density of states functions is necessary and not the exact expression. For 3D metal, $Z(E) \propto \sqrt{E}$ so that

$$Z(E) = \alpha\sqrt{E}, \quad \text{where } \alpha \text{ is a constant}$$

Specific heat capacity – Form of $Z(E)$

- At $T = 0$ K, electron has energy less than or equal to Fermi energy level.
- The number of electrons per unit volume n_c at $T = 0$ K is given by

$$\begin{aligned} n_c &= \int_0^{E_F} Z(E) P(E) dE \\ &= \int_0^{E_F} Z(E) dE \\ &= \int_0^{E_F} \alpha \sqrt{E} dE \\ &= \frac{2}{3} \alpha E_F^{3/2} \end{aligned}$$

$$\therefore n_c = \frac{2}{3} \alpha E_F^{3/2}.$$

Specific heat capacity – Conduction electron density n_c

- At $T \neq 0$ K, due to thermalization, some of the electrons vacate the lower energy levels and occupy the higher energy levels.
- The number of electrons per unit volume n_c at $T \neq 0$ K is given by

$$\begin{aligned} n_c &= \int_0^{\infty} Z(E)P(E) dE \\ &= \int_0^{\infty} \frac{\alpha \sqrt{E}}{\exp\left(\frac{E-E_F}{k_B T}\right) + 1} dE \end{aligned}$$

Note that integration limits are 0 to ∞ !

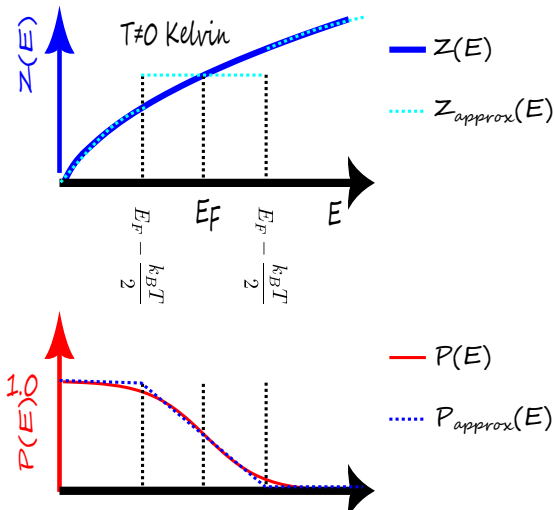
- We observe that because of the strong variation of Fermi Dirac distribution only near the Fermi level, the electron with energy belonging to range $E_F - \frac{k_B T}{2}$ to E_F jumps to energy in the range E_F to $E_F + \frac{k_B T}{2}$. Let us use this to **approximate** the integral.

Key Insight



Electrons with energy very much less than Fermi energy level do not participate in the thermalization process.

Specific heat capacity – Approximations to $Z(E)$, $P(E)$

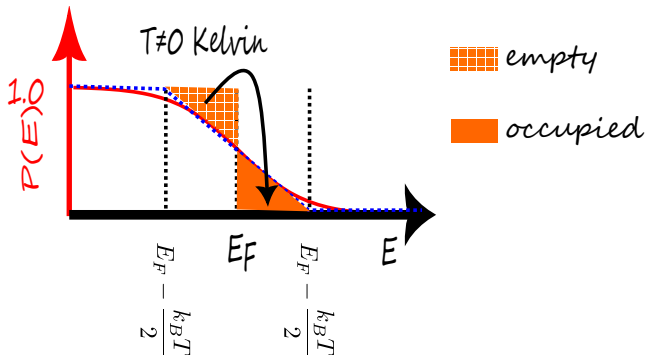


- Let us approximate the density of states in the neighbourhood of E_F as

$$Z(E) \simeq Z(E_F)$$

- Let us approximate probability distribution as piece-wise straight line function.

Specific heat capacity – Fraction of thermalized electrons



- The number of thermalized electrons is equal to area of the filled triangle multiplied by the density of states in the neighbourhood.
- Area of the filled triangle is

$$A = \frac{1}{2} \frac{k_B T}{2} \frac{1}{2} = \frac{1}{8} k_B T$$

$$\therefore n_{\text{thermal}} = \frac{1}{8} k_B T \cdot Z(E_F) = \frac{1}{8} \alpha k_B T \sqrt{E}$$

Specific heat capacity – Fraction of thermalized electrons

- The fraction of thermalized electrons is given by ratio of thermalized electrons to total electrons

$$\begin{aligned} f_{\text{thermalized}} &= \frac{n_{\text{thermal}}}{n_c} \\ &= \frac{1}{8} k_B T \frac{\alpha \sqrt{E}}{\frac{2}{3} \alpha E_F^{3/2}} \\ &= \frac{3}{16} \frac{k_B T}{E_F} \end{aligned}$$

Problem

The Fermi level of silver is 5.5 eV. Calculate the fraction of free electrons at room temperature.

Key Insight

The fraction of thermalized electrons is of the order of $\frac{k_B T}{E_F}$.

Specific heat capacity – Energy of thermalized electron

- The net increase in energy of thermalized electron is

$$\Delta E_{\text{thermalized}} = E_F + \frac{k_B T}{2} - \left(E_F - \frac{k_B T}{2} \right) = k_B T$$

- Since only a fraction of the electrons are thermalized, the net increase in energy of 1 electron is

$$\begin{aligned}\Delta E_{1 \text{ el}} &= \Delta E_{\text{thermalized}} \cdot f_{\text{thermalized}} \\ &= \frac{3}{16} k_B T \cdot \frac{k_B T}{E_F}\end{aligned}$$

- Therefore, the net increase in energy of 1 mole of electrons is

$$\begin{aligned}\Delta E_{1 \text{ mole}} &= \Delta E_{1 \text{ el}} \cdot N_A \\ &= \frac{3}{16} (k_B N_A) T \cdot \frac{k_B T}{E_F} \\ &= \frac{3}{16} R \frac{k_B T^2}{E_F}\end{aligned}$$

Specific heat capacity – QFET

- The heat capacity is given by

$$\begin{aligned}C_{\text{theory}}^{\text{quantum}} &= \frac{dE}{dT} \\&= \frac{3}{8} R \frac{k_B T}{E_F} \\&= \frac{3}{2} R \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right) \\&= C_{\text{theory}}^{\text{classical}} \cdot \left(\frac{1}{4} \frac{k_B T}{E_F} \right)\end{aligned}$$

$$\therefore C_{\text{theory}}^{\text{quantum}} \simeq C_{\text{theory}}^{\text{classical}} \cdot \left(\frac{k_B T}{E_F} \right).$$

- Electronic contribution to specific heat at low T is **linear**.
- Phonons i.e. lattice vibrations also contribute to specific heat capacity C_{phonon} that has cubic form.

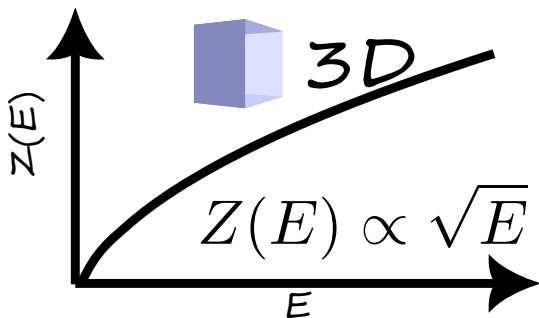
$$\begin{aligned}\therefore C_{\text{theory}}^{\text{quantum}} &= C_{\text{electron}} + C_{\text{phonon}} \\&= AT + BT^3\end{aligned}$$

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Density of states (3D)

- The form of the DOS in 3D is $Z(E) \propto \sqrt{E}$

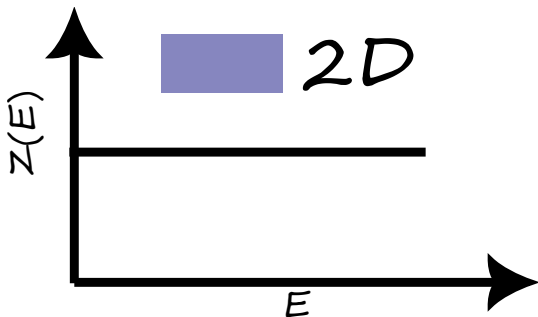


Problem

If $Z_{3D}(E) = k\sqrt{E}$, where k is a constant, find the expression for conduction electron number density n_c in terms of Fermi level E_F at $T = 0\text{ K}$ in a 3D material.

Density of states (2D)

- The form of the DOS in 2D is $Z(E) = \text{const}$



Problem

If $Z_{2D}(E) = k$, where k is a constant, find the expression for conduction electron number density n_c in terms of Fermi level E_F at $T = 0\text{ K}$ in a 2D material.

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Merits and demerits of QFET

Merits

- Electrical conductivity
- Thermal conductivity
- Heat capacity

Demerits

- Cannot explain classification of condensed matter into metals, semiconductors and insulators
- Occurrence of positive Hall coefficient in some metals like Zn.

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$E - k$ diagram – 1D case

- In a 1D metal, the electron is free as well as independent.
- For a free electron, the energy is related to the momentum p by

$$E = \frac{p^2}{2m}.$$

- However, since the electron behaves also as a wave, its wavelength λ is related to the momentum by the de Broglie relation

$$\lambda = \frac{h}{p} \quad \Rightarrow \quad p = \frac{h}{\lambda}.$$

- By definition, the wavevector k of the electron is inversely related to the wavelength as

$$k = \frac{2\pi}{\lambda}$$

- Therefore, the energy is related to the wavevector by

$$E = \frac{1}{2m} \left(\frac{h}{\lambda} \right)^2 = \frac{1}{2m} \left(\frac{hk}{2\pi} \right)^2 = \frac{\hbar^2 k^2}{2m}.$$

$E - k$ diagram – 3D case

- The analysis for 1D case can be extended to the 3D case by considering the components of momentum p_x, p_y, p_z

$$\vec{p} = p_x \hat{i} + p_y \hat{j} + p_z \hat{k}$$

so that

$$E = \frac{p_x^2 + p_y^2 + p_z^2}{2m}.$$

- The wavevector has three components k_x, k_y, k_z so that

$$\vec{k} = k_x \hat{i} + k_y \hat{j} + k_z \hat{k}.$$

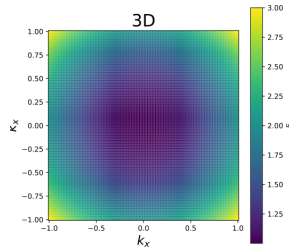
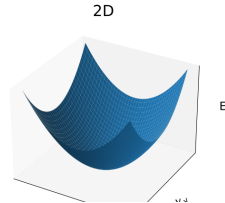
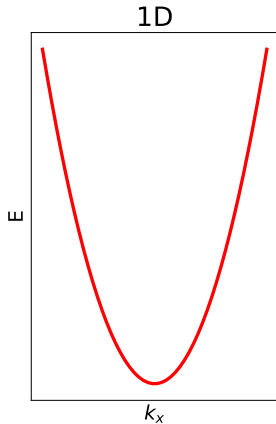
- The energy of the free electron is given by

$$E = \frac{\hbar^2}{2m}(k_x^2 + k_y^2 + k_z^2)$$

Problem

Fermi wavevector k_F If the Fermi energy is $E_F = 5 \text{ eV}$, find the Fermi wavevector.

Form of $E - k$ diagram – 1D, 2D, 3D



Key Insight

Form of $E - k$ is parabolic.



Summary of Quantum free electron theory (QFET)

- Derived density of states function $Z(E)$.
- Quantum free electron theory addressed the electronic contribution to specific heat of metals
- However, the theory has drawbacks. Some of them are
 - Cannot explain anomalous sign of Hall coefficient in some metals.
 - Cannot explain classification of materials into conductors, semi-conductors and insulators.

Reason for drawbacks

- Invalidity of free electron approximation.
- Drawbacks addressed by Bloch's quantum band theory.